



Test Design Specifications

Online Fact-Checking System

(Sebenarnya.my)

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Date: 4 January 2026

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Design Specification ID: SeBenarnya_TDS_1.0.0	

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1.0	4 January 2026	All	Original Document

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1.0 Introduction

1.1 Purpose

This test design specification supports the following objectives:

- To identify the test items covered in the **SeBenarnya Online Fact-Checking System**.
- To further refine the test approach based on the test plan.
- To define the sources of information used to prepare the test design.
- To identify the general test cases to be executed for the selected features.

1.2 Scope

This test design specification covers only selected features of the **SeBenarnya Online Fact-Checking System**, which are:

- AI-Based News Classification
- Multi-Language Detection and Verification
- User Behaviour Analytics and Reputation System

1.3 References

This document is prepared in reference to the following documents:

1. ATMS Test Design Specification. (2013, April 16). Malaysian Software Testing Board.

The following documents provide the test basis for the test design:

2. SeBenarnya Software Requirements Specifications (Sebenarnya.my_SRS_1.0.0)
3. SeBenarnya Test Plan (Sebenarnya.my_TP_1.0.0)

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2.0 Test Design

2.1 Features to be Tested

The following table contains the features to be tested based on SeBenarnya Software Requirement Specification (SRS) for this iteration. Listed together are the function ID and its corresponding functions and estimated risk level.

Function ID	Functions	Risk Level
F001	Multi-Language Detection	High
F002	Language Routing to Appropriate Classifier	High
F003	Cross-Language Duplicate Detection	High
F004	Translation Validation	High
F005	AI-Based News Classification	High
F006	Multi-Language Reporting	Medium
F007	User Submission	Medium
F008	Fact-Checker Verification Workflow	Medium

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2.2 Approach Refinements

As discussed in Test Plan (SeBenarnya_TP_1.0.0), test cases for Sebenarnya system will be designed using five techniques which are covered in this section. Techniques are applied based on suitability of techniques according to the nature of features.

The techniques that will be applied are as follows:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.1 F001 Multi-Language Detection

Five techniques are identified for designing tests for this feature, including:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.1.1 Equivalence Partitioning for Language Detection

Equivalence partitioning is used to split Language Detection into possible partitions for testing.

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Input of News Content Language

Invalid Partition 1	Valid Partition 1	Valid Partition 2	Valid Partition 3	Invalid Partition 2
Empty Content	Malay Language Content	English Language Content	Mandarin Language Content	Mixed / Unidentifiable Language

Table 2.2.1a Language Detection Equivalence Partitions

Equivalent Partitioning of Language Detection = { empty, Malay, English, Mandarin, mixed/unidentifiable }

Language Detection Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-01-001	News Content = empty	TCOV-01-001	Invalid partition 1	"" (empty string)
TCON-01-002	Pure Malay language content	TCOV-01-002	Valid partition 1	"Kerajaan mengumumkan dasar baharu untuk pendidikan"
TCON-01-003	Pure English language content	TCOV-01-003	Valid partition 2	"Government announces new education policy today"
TCON-01-004	Pure Mandarin language content	TCOV-01-004	Valid partition 3	"政府今天宣布新的教育政策"
TCON-01-005	Mixed language content (code-switching)	TCOV-01-005	Invalid partition 2	"Today kerajaan announce policy baharu"

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TCOV-01-00 6	Content with special characters only	TCOV-01-00 6	Invalid partition 1	"@#\$\$%^&*()"
TCOV-01-00 7	Content with numbers only	TCOV-01-00 7	Invalid partition 1	"123456789"
TCOV-01-00 8	Malay content with English names	TCOV-01-00 8	Valid partition 1	"Perdana Menteri Datuk Seri Anwar Ibrahim berucap hari ini"

Table 2.2.1.1b Language Detection Equivalence Partitioning Test Conditions & Coverage

2.2.1.2 Boundary Value Analysis

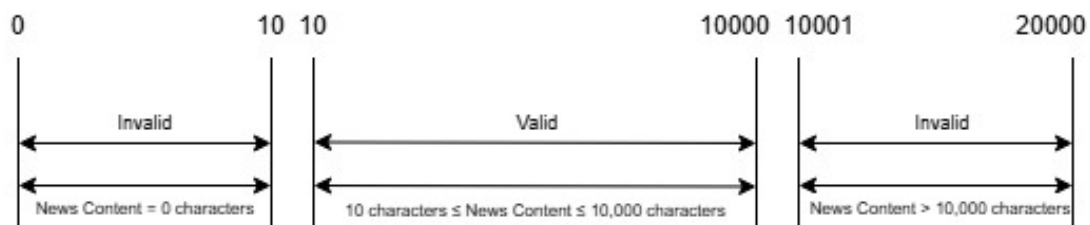
Boundary Value Analysis is used to test the boundary values of text length for language detection accuracy.

Boundary Value Assumptions

- Minimum length for reliable language detection = 10 characters
- Maximum length for processing = 10,000 characters

Equivalence Partitions

- EP1: $0 \leq \text{Content} < 10$ characters (insufficient for detection)
- EP2: $10 \text{ characters} \leq \text{Content} \leq 10,000$ characters (valid)
- EP3: $\text{Content} > 10,000$ characters (exceeds limit)



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Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-01-009	Content = 0 characters	TCOV-01-009	Content = 0 characters (Invalid)	""
TCON-01-010	Content = 9 characters	TCOV-01-010	Content < 10 characters (Invalid)	"Berita"
TCON-01-011	Content = 10 characters	TCOV-01-011	Content = 10 characters (Valid)	"Berita ini"
TCON-01-012	Content = 5,000 characters	TCOV-01-012	Content = mid-range (Valid)	Long article text
TCON-01-013	Content = 10,000 characters	TCOV-01-013	Content = maximum (Valid)	Maximum length article
TCON-01-014	Content = 10,001 characters	TCOV-01-014	Content > maximum (Invalid)	Article exceeding limit

2.2.1.3 Decision Table Testing

Decision table testing is used to test valid combinations of language characteristics and detection outcomes.

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Test Conditions:

- TCON-01-002: Pure Malay language content
- TCON-01-003: Pure English language content
- TCON-01-004: Pure Mandarin language content
- TCON-01-005: Mixed language content

Condition	TCOV-01-002	TCOV-01-003	TCOV-01-004	TCOV-01-005
Contains Malay words	Y	N	N	Y
Contain English words	N	Y	N	Y
Contain Mandarin characters	N	N	Y	Y
Single dominant language	Y	Y	Y	N
Output	Malay	English	Mandarin	Error/Primary Language

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2.2.1.4 State Transition Testing

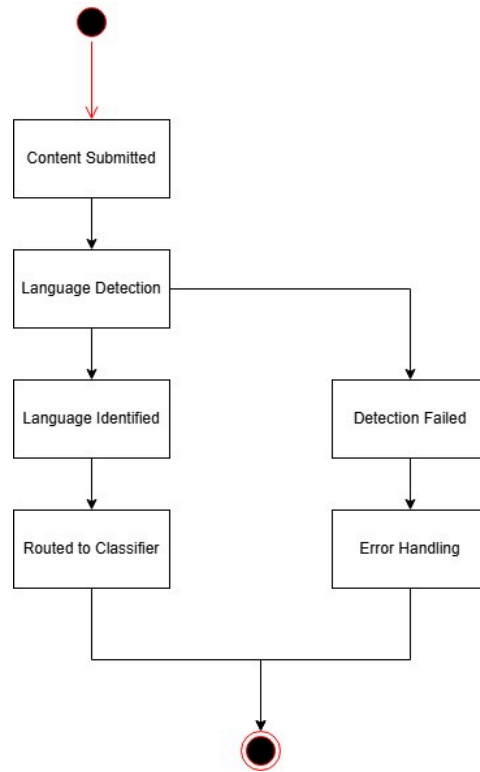


Figure 2.2.1.4a Multi-Language Detection State Diagram

State Table

State/Input	Submit Content	Detect Language	Language Confirmed	Route to Classifier
Submitted	S1/TCOV-01-015	-	-	-
Detecting	-	S2/TCOV-01-016	-	-
Identified	-	-	S3/TCOV-01-017	-
Routed	-	-	-	S4/TCOV-01-018

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State Definitions:

- S1: Content submitted for language detection
- S2: System analyzing language patterns
- S3: Language successfully identified
- S4: Content routed to appropriate classifier

State Transition Coverage:

Test Coverage ID	Test Coverage
TCOV-01-015	S1 to S2 with Submit Content
TCOV-01-016	S2 to S3 with Detect Language
TCOV-01-017	S3 to S4 with Language Confirmed
TCOV-01-018	S4 completion with Route to Classifier

2.2.1.5 Use Case Testing

Multi-Language Detection [REQ-02-001]

Use Case	UC001	Version	1.0
Feature	F001		

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Purpose	To automatically detect the language of submitted news content with high accuracy
Actor	System, User
Trigger	User submits news content for verification
Precondition	User must submit readable news content. Fact-checker must be logged into the system with appropriate access rights.
Main Flow	1. User submits news content through the system interface 2. System receives the submitted content 3. System analyzes linguistic patterns and characteristics 4. System identifies the primary language (Malay, English, or Mandarin) 5. System stores the detected language metadata 6. System proceeds to language-specific routing 7. Use case ends
Alternative Flow	None.
Exception Flow	E1: Insufficient Content for Detection [REQ-02-001] 1. System detects content length is below minimum threshold 2. System displays error message requesting longer content

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	3. Use case ends E2: Unidentifiable Language [REQ-02-001] 1. System cannot confidently identify the language 2. System flags content for manual review 3. System notifies user of detection failure 4. Use case ends
Rules	● R1: Content must have minimum 10 characters for reliable detection ● R2: System must achieve minimum 95% accuracy for single-language content ● R3: Mixed-language content should identify the dominant language or flag for review
Note	None
Author	Nur Izzah Dania binti Hipni Shahlizal

Use Case Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCOV-01-019	Main Flow	TCOV-01-019	Main flow	Submit valid Malay content

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TCOV-01-020	Exception Flow – Insufficient Content	TCOV-01-020	Exception flow	Submit 5-character content
TCOV-01-021	Exception Flow – Unidentifiable Language	TCOV-01-021	Exception flow	Submit mixed/garbled text

2.2.2 F002 Language Routing to Appropriate Classifier

Five techniques are identified for designing tests for this feature, including:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.2.1 Equivalence Partitioning for Language Routing

Equivalence partitioning is used to split Language Detection into possible partitions for testing.

Input of Detected Language

Invalid Partition 1	Valid Partition 1	Valid Partition 2	Valid Partition 3	Invalid Partition 2
No language detected	Malay detected	English detected	Mandarin detected	Unsupported language

Table 2.2.2.1a Language Routing Equivalence Partitions

Equivalent Partitioning of Language Routing = { empty, Malay, English, Mandarin, unsupported }

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Language Detection Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-02-001	No language detected	TCOV-02-001	Invalid partition 1	null language value
TCON-02-002	Malay language detected	TCOV-02-002	Valid partition 1	language = "ms"
TCON-02-003	English language detected	TCOV-02-003	Valid partition 2	language = "en"
TCON-02-004	Mandarin language detected	TCOV-02-004	Valid partition 3	language = "zh"
TCON-02-005	Unsupported language detected	TCOV-02-005	Invalid partition 2	language = "fr" (French)
TCON-02-006	Routing with valid classifier availability	TCOV-02-006	Valid partition 1	Malay classifier active
TCON-02-007	Routing with classifier unavailable	TCOV-02-007	Invalid partition 2	Classifier offline

Table 2.2.2.1b Language Routing Equivalence Partitioning Test Conditions & Coverage

2.2.2.2 Boundary Value Analysis

Boundary Value Analysis is used to test routing success rate thresholds.

Boundary Value Assumptions

- Minimum confidence threshold for routing = 70%

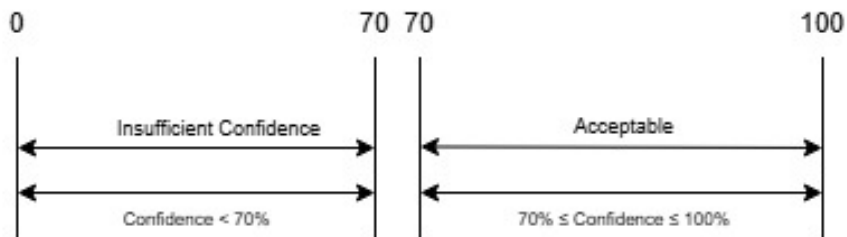
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- Maximum confidence threshold = 100%

Equivalence Partitions

- EP1: Confidence < 70% (insufficient confidence)

- EP2: $70\% \leq \text{Confidence} \leq 100\%$ (acceptable)



Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCOV-02-008	Confidence = 0%	TCOV-02-008	Below minimum (Invalid)	Detection failed
TCOV-02-009	Confidence = 69%	TCOV-02-009	Just below threshold (Invalid)	Uncertain detection
TCOV-02-010	Confidence = 70	TCOV-02-010	At minimum threshold (Valid)	Acceptable detection
TCOV-02-011	Confidence = 95%	TCOV-02-011	High confidence (Valid)	Strong detection

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TCOV-02-012	Confidence = 100%	TCOV-02-012	Maximum confidence (Valid)	Perfect detection
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2.2.2.3 Decision Table Testing

Condition	TCOV-02-002	TCOV-02-003	TCOV-02-004	TCOV-02-005	TCOV-02-007
Language Detected	Malay	English	Mandarin	Unsupported	Malay
Confidence $\geq$ 70%	Y	Y	Y	Y	Y
Classifier available	Y	Y	Y	N/A	N
Output	Route to Malay Classifier	Route to English Classifier	Route to Mandarin Classifier	Error: Unsupported	Error: Classifier Unavailable

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2.2.2.4 State Transition Testing

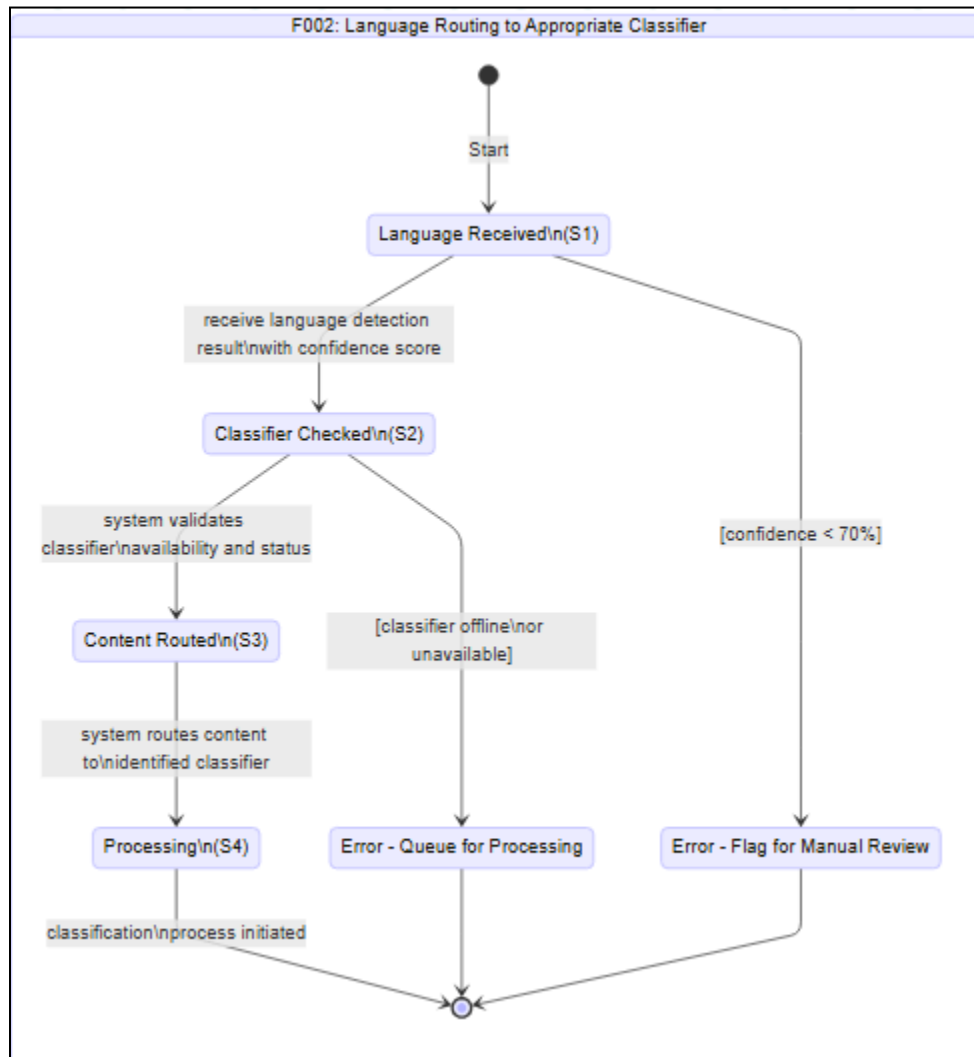


Figure 2.2.2.4a State Diagram

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State Table

State/Input	Receive Language	Check Classifier	Route Content	Process Complete
Submitted	S1/TCOV-02-013	-	-	-
Detecting	-	S2/TCOV-02-014	-	-
Identified	-	-	S3/TCOV-02-015	-
Routed	-	-	-	S4/TCOV-02-016

State Definitions:

- S1: Language detection result receive
- S2: System verifies appropriate classifier availability
- S3: Content routed to language-specific classifier
- S4: Classifier processing initiated

State Transition Coverage:

Test Coverage ID	Test Coverage
TCOV-02-013	S1 to S2 with Receive Language
TCOV-02-014	S2 to S3 with Check Classifier
TCOV-02-015	S3 to S4 with Route Content
TCOV-02-016	SS4 completion with Process Complete

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2.2.2.5 Use Case Testing

Language Routing to Appropriate Classifier [REQ-02-002]

Use Case	UC002	Version	1.0
Feature	F002		
Purpose	To route detected language content to the correct language-specific classifier module		
Actor	System		
Trigger	Language detection completed successfully		
Precondition	Language must be detected with confidence $\geq 70\%$; Appropriate classifier must be available.		
Main Flow	1. System receives language detection result with confidence score 2. System validates confidence score meets minimum threshold (70%) 3. System identifies appropriate language-specific classifier (Malay/English/Mandarin) 4. System verifies classifier availability and status 5. System routes content to the identified classifier 6. System initiates classification process 7. Use case ends		

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Alternative Flow	None.
Exception Flow	E1: Low Confidence Score [REQ-02-002] 1. System detects confidence score below 70% 2. System flags content for manual language verification

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	3. System notifies administrator 4. Use case ends E2: Classifier Unavailable [REQ-02-002] 1. System detects target classifier is offline or unavailable 2. System queues content for processing when classifier becomes available 3. System notifies administrator 4. Use case ends
Rules	● R1: Only content with confidence $\geq 70\%$ should be automatically routed ● R2: System must verify classifier availability before routing ● R3: Failed routing attempts must be logged and queued for retry
Note	None
Author	Nur Izzah Dania binti Hipni Shahlizal

Use Case Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-02-017	Main Flow	TCOV-02-017	Main flow	Route Malay content with 95% confidence

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TCOV-02-018	Exception flow	TCOV-02-018	Exception flow	Route with 65% confidence
TCOV-02-019	Exception flow	TCOV-02-019	Exception flow	Route when classifier offline

2.2.3 F003 Cross-Language Duplicate Detection

Five techniques are identified for designing tests for this feature, including:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.3.1 Equivalence Partitioning for Duplicate Detection

Input of Detected Language

Invalid Partition 1	Valid Partition 1	Valid Partition 2	Valid Partition 3	Invalid Partition 2
Empty Comparison set	Unique content (no duplicates)	Exact duplicates in same language	Semantic duplicates across languages	Corrupted comparison data

Table 2.2.3.1a Cross-Language Duplicate Detection Equivalence Partitions

Equivalent Partitioning of Duplicate Detection = { empty, unique, same-language duplicate, cross-language duplicate, corrupted }

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Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-03-001	No existing content to compare	TCOV-03-001	Invalid partition 1	Empty database
TCON-03-002	Completely unique content	TCOV-03-002	Valid partition 1	New unique news story
TCON-03-003	Exact duplicate in same language	TCOV-03-003	Valid partition 2	Word-for-word copy
TCON-03-004	Semantic duplicate in different language	TCOV-03-004	Valid partition 3	Same story in English and Malay)
TCON-03-005	Similar but not duplicate content	TCOV-03-005	Valid Partition 1	Related but different news
TCON-03-006	Paraphrased content across languages	TCOV-03-006	Valid partition 1	Rephrased story in Mandarin
TCON-03-007	Partial duplicate content	TCOV-03-007	Valid partition 2	Article with shared paragraphs

Table 2.2.3.1b Cross-Language Duplicate Detection Equivalence Partitioning Test Conditions & Coverage

2.2.3.2 Boundary Value Analysis

Boundary Value Analysis for similarity threshold detection.

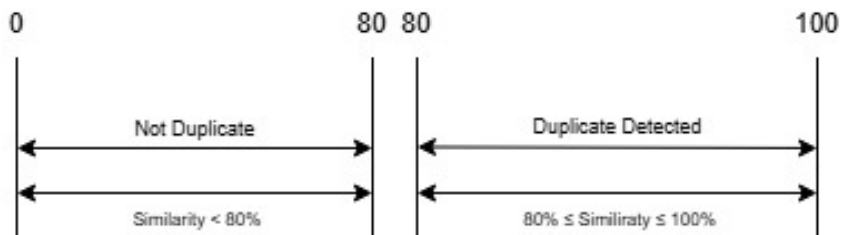
Boundary Value Assumptions

- Minimum similarity threshold for duplicate = 80%
- Maximum similarity = 10

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Equivalence Partitions

- EP1: Similarity < 80% (not duplicate)
- EP2: $80\% \leq \text{Similarity} \leq 100\%$ (duplicate detected)



Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-03-008	Similarity = 0%	TCOV-03-008	Completely different (Valid - no duplicate)	Unrelated content
TCON-03-009	Similarity = 79	TCOV-03-009	Just below threshold (Valid - no duplicate)	Similar but distinct
TCON-03-010	Similarity = 80%	TCOV-03-010	At threshold (Valid - duplicate detected)	Minimum duplicate match
TCON-03-011	Similarity = 95%	TCOV-03-011	High similarity (Valid - duplicate)	Strong duplicate match

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TCOV-03-012	Similarity = 100%	TCOV-03-012	Exact match (Valid - duplicate)	Perfect duplicate
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2.2.3.3 Decision Table Testing

Condition	TCOV-03-002	TCOV-03-003	TCOV-03-004	TCOV-03-006	TCOV-03-005
Content exists in database	N	Y	Y	Y	Y
Same Language	N/A	Y	N	N	Y
Similarity $\geq$ 80%	N/A	Y	Y	Y	N
Semantic meaning matches	N/A	Y	Y	Y	N
Output	No Duplicate (Proceed)	Duplicate Detected (Same Language)	Cross-Language Duplicate Detected	Cross-Language Duplicate Detected	No Duplicate (Proceed)

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2.2.3.4 State Transition Testing

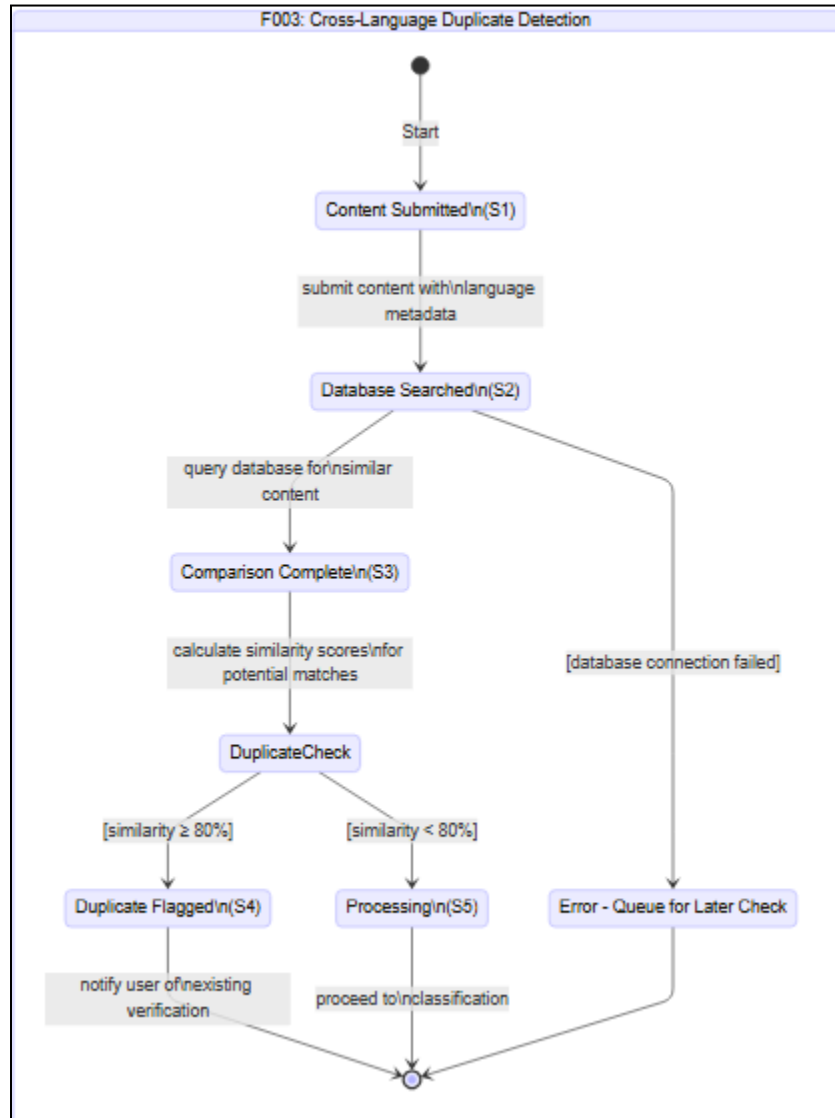


Figure 2.2.3.4a State Diagram

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State Table

State/Input	Submit Content	Search Database	Compare Semantics	Flag Duplicate	Process Content
Content Submitted	S1/TCOV-03-013	-	-	-	
Database Searched	-	S2/TCOV-03-014	-	-	
Comparison Complete	-	-	S3/TCOV-03-015	-	
Duplicate Flagged	-	-	-	S4/TCOV-03-016	
Processing	-	-	-	-	S5/TCOV-03-017

State Definitions:

- S1: New content submitted for duplicate checking
- S2: System searches database for similar content
- S3: Semantic comparison completed across languages
- S4: Duplicate identified and flagged
- S5: Content proceeds to classification or is rejected

State Transition Coverage:

Test Coverage ID	Test Coverage
TCOV-03-013	S1 to S2 with Submit Content
TCOV-03-014	S2 to S3 with Search Database

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TCOV-03-015	S3 to S4 with Compare Semantics
TCOV-03-016	S4 to S5 with Flag Duplicate
TCOV-03-017	S5 completion with Process Content

2.2.3.5 Use Case Testing

Cross-Language Duplicate Detection [REQ-02-003]

Use Case	UC003	Version	1.0
Feature	F003		
Purpose	To detect duplicate or highly similar news content across different languages using semantic comparison		
Actor	System		
Trigger	Content successfully routed to classifier		
Precondition	Content must have detected language; Database must contain existing verified content for comparison		
Main Flow	1. System receives content with language metadata 2. System queries database for similar content in all supported languages 3. System performs cross-language semantic similarity analysis		

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	- System calculates similarity scores for potential matches - System identifies duplicates where similarity $\geq 80\%$ - System flags duplicate content with reference to original submission - System notifies user of existing verification result - Use case ends
Alternative Flow	A1: No Duplicate Found - At step 5, system finds no matches above similarity threshold - System marks content as unique - System proceeds with classification process - Use case ends
Exception Flow	**E1: Database Connection Failed [REQ-02-003]** - System cannot access database for comparison - System logs error and notifies administrator - System queues content for later duplicate check - Use case ends
Rules	- R1: Similarity threshold must be $\geq 80\%$ to flag as duplicate - R2: System must check across all supported languages (Malay, English, Mandarin) - R3: Duplicate detection must preserve original submission metadata

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Note	None
Author	Nur Izzah Dania binti Hipni Shahlizal

Use Case Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-03-018	Main Flow	TCOV-03-018	Main flow	Submit content with existing Mandarin duplicate
TCON-03-019	Alternative Flow - No Duplicate	TCOV-03-019	Alternative flow	Submit unique content
TCON-03-020	Exception Flow – Database Error	TCOV-03-020	Exception flow	Simulate database failure

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2.2.4 F004 Translation Validation

Five techniques are identified for designing tests for this feature, including:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.4.1 Equivalence Partitioning for Translation Quality

Input of Detected Language

Invalid Partition 1	Valid Partition 1	Valid Partition 2	Invalid Partition 2
No translation available	Accurate translation preserving meaning	Acceptable translation with minor variations	Poor quality translation with meaning loss

Table 2.2.4.1a Translation Validation Equivalence Partitions

Equivalent Partitioning of Translation Quality = { none, accurate, acceptable, poor }

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCOV-04-00 1	Translation not available	TCOV-04-00 1	Invalid partition 1	Translation service offline
TCOV-04-00 2	High quality accurate translation	TCOV-04-00 2	Valid partition 1	Perfect meaning preservation
TCOV-04-00 3	Acceptable quality translation	TCOV-04-00 3	Valid partition 2	Minor contextual differences

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TCOV-04-00 4	Poor quality translation	TCOV-04-00 4	Invalid partition 2	Meaning distorted or lost
TCOV-04-00 5	Translation with technical terms preserved	TCOV-04-00 5	Valid partition 1	Medical/legal terms correctly translated
TCOV-04-00 6	Translation with idioms/phrases	TCOV-04-00 6	Valid partition 2	Culturally adapted expressions
TCOV-04-00 7	Translation with factual errors	TCOV-04-00 7	Invalid partition 2	Numbers, dates, names incorrect

Table 2.2.4.1b Translation Validation Equivalence Partitioning Test Conditions & Coverage

2.2.4.2 Boundary Value Analysis

Boundary Value Analysis for translation quality score.

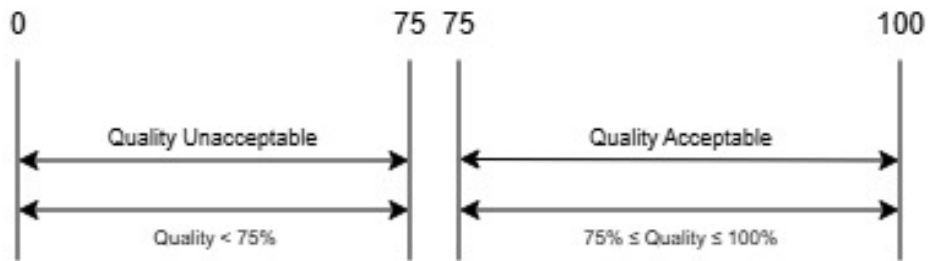
Boundary Value Assumptions

- Minimum acceptable quality score = 75%
- Maximum quality score = 100%

Equivalence Partitions

- EP1: Quality < 75% (unacceptable)
- EP2: 75% ≤ Quality ≤ 100% (acceptable)

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Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-04-008	Quality = 0%	TCOV-04-008	Translation failed (Invalid)	Gibberish output
TCON-04-009	Quality = 74%	TCOV-04-009	Just below threshold (Invalid)	Poor translation
TCON-04-010	Quality = 75	TCOV-04-010	At minimum threshold (Valid)	Minimally acceptable
TCON-04-011	Quality = 90%	TCOV-04-011	High quality (Valid)	Good translation
TCON-04-012	Quality = 100%	TCOV-04-012	Perfect quality (Valid)	Flawless translation

2.2.4.3 Decision Table Testing

Condition	TCOV-04-002	TCOV-04-003	TCOV-04-004	TCOV-04-007
Translation available	Y	Y	Y	Y

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Quality score $\geq$ 75%	Y	Y	N	N
Meaning preserved	Y	Y	N	N
Factual accuracy maintained	Y	Y	Y	N
Output	Approved (Proceed to Classification)	Approved with Review Flag	Rejected (Retranslate)	Rejected (Factual Error)

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2.2.4.4 State Transition Testing

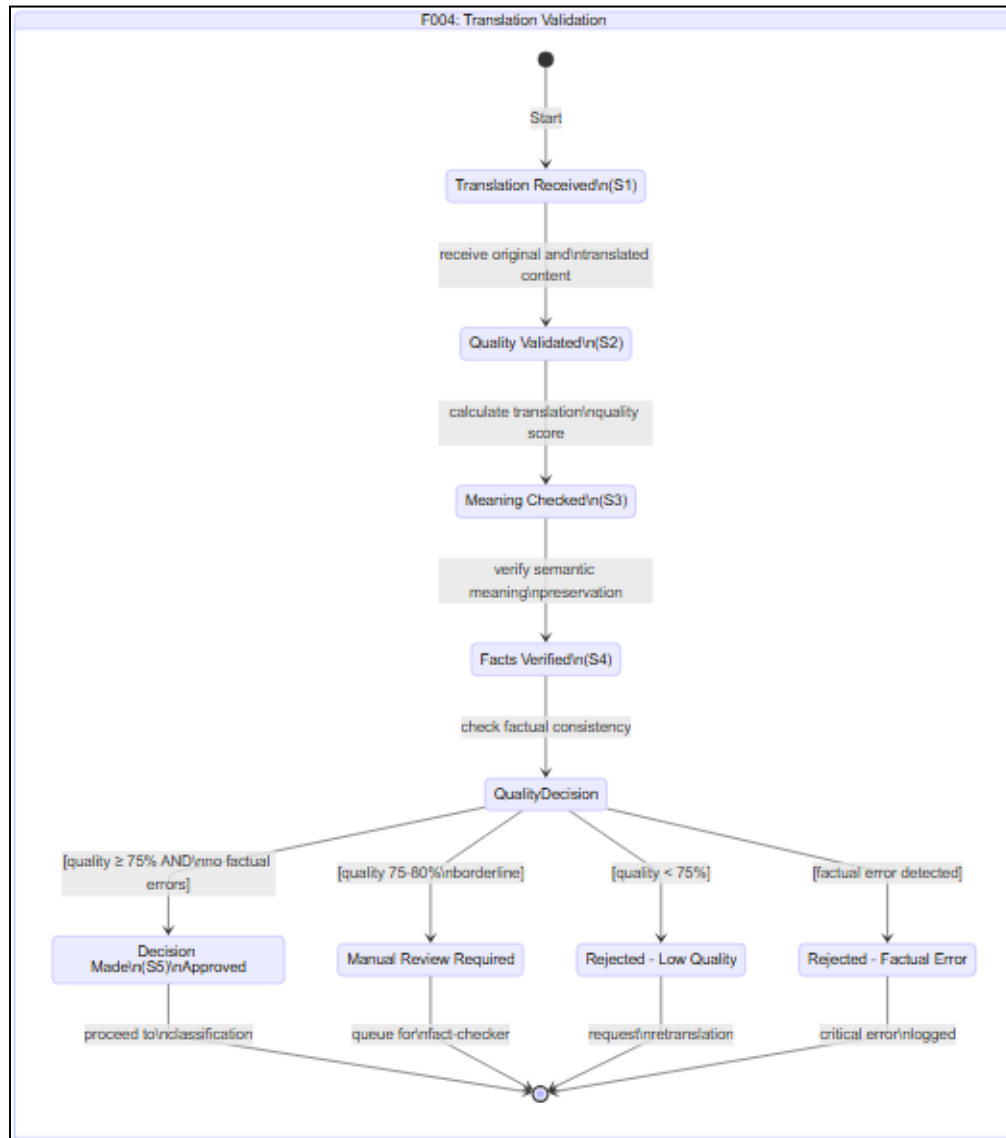


Figure 2.2.4.4a State Diagram

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Design Specification ID: SeBenarnya_TDS_1.0.0	

State Table

State/Input	Receive Translation	Validate Quality	Check Meaning	Verify Facts	Approve/Reject
Translation Received	S1/TCOV-04-013	-	-	-	
Quality Validated	-	S2/TCOV-04-014	-	-	
Meaning Checked	-	-	S3/TCOV-04-015	-	
Facts Verified	-	-	-	S4/TCOV-04-016	
Decision Made	-	-	-	-	S5/TCOV-04-017

State Definitions:

- S1: Translation output received from translation service
- S2: System validates translation quality score
- S3: System checks semantic meaning preservation
- S4: System verifies factual accuracy of translated content
- S5: Translation approved or rejected for classification

State Transition Coverage:

Test Coverage ID	Test Coverage
TCOV-04-013	S1 to S2 with Receive Translation

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TCOV-04-014	S2 to S3 with Validate Quality
TCOV-04-015	S3 to S4 with Check Meaning
TCOV-04-016	S4 to S5 with Verify Facts
TCOV-04-017	S5 completion with Approve/Reject

2.2.4.5 Use Case Testing

Translation Validation [REQ-02-004]

Use Case	UC004	Version	1.0
Feature	F004		
Purpose	To validate translation accuracy and quality before performing classification		
Actor	System		
Trigger	Content requires translation for cross-language analysis or verification		
Precondition	Original content and translated content must be available; Translation service must be operational		
Main Flow	1. System receives original content and its translation		

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	- System calculates translation quality score using AI validation model - System verifies semantic meaning preservation between original and translation - System checks factual consistency (numbers, dates, names, locations) - System validates cultural context appropriateness - System evaluates technical terminology accuracy - System approves translation if quality score $\geq 75\%$ - System proceeds to classification with validated translation - Use case ends
Alternative Flow	A1: No Duplicate Found - At step 7, quality score is between 75-80% (borderline) - System flags translation for human reviewer - System queues content for manual validation - System notifies fact-checker - Use case ends
Exception Flow	E1: Translation Quality Insufficient [REQ-02-004] - At step 7, quality score $< 75\%$ - System rejects translation - System requests retranslation with improved parameters - System logs translation failure

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	5. Use case ends E2: Factual Error Detected [REQ-02-004] 1. At step 4, system detects factual inconsistency 2. System flags critical error 3. System rejects translation regardless of quality score 4. System notifies administrator 5. Use case ends
Rules	● R1: Minimum quality threshold is 75% for automatic approval ● R2: Factual errors must cause automatic rejection ● R3: Technical and medical terms must be translated with 100% accuracy ● R4: Cultural context must be preserved or appropriately adapted
Note	None
Author	Nur Izzah Dania binti Hipni Shahlizal

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Use Case Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-04-018	Main Flow	TCOV-04-018	Main flow	Validate high-quality translation (90%)
TCON-04-019	Alternative Flow - Manual Review	TCOV-04-019	Alternative flow	Validate borderline translation (77%)
TCON-04-020	Exception Flow – Low Quality	TCOV-04-020	Exception flow	Validate poor translation (60%)
TCON-04-021	Exception Flow – Factual Error	TCOV-04-021	Exception flow	Validate translation with wrong date

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2.2.5 F005 AI-Based News Classification

Five techniques are identified for designing test for this feature, including:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.5.1 Equivalence Partitioning for News Content

Equivalence partitioning is used to split News Content verification into possible partitions for testing.

$1 \text{ character} \leq \text{News Content} \leq 10,000 \text{ characters}$

Input of News Content

Invalid Partition 1	Valid Partition 1	Invalid Partition 2
News Content = 0 character	Readable and meaningful news content	Unsupported, manipulated, or meaningless content

Table 2.2.5.1a News Content Equivalence Partitions

Equivalent Partitioning of News Content = { empty, valid readable content, invalid / manipulated content }

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News Content Equivalence Partitioning Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCOV-05-001	News Content = 0 character	TCOV-05-001	Invalid partition 1	(empty input)
TCOV-05-002	Valid readable news content	TCOV-05-002	Valid partition 1	“Breaking news: Health ministry denies the rumour.”
TCOV-05-003	News content updated after fact-checker verification	TCOV-05-003	Valid partition 1	Fact-checker corrects classification
TCOV-05-004	Rephrased content similar to known fake news	TCOV-05-004	Valid partition 1	Paraphrased misinformation article
TCOV-05-007	Adversarially manipulated news content	TCOV-05-007	Invalid partition 2	“v1rus spread g0vernment coverup”
TCOV-05-008	News content containing sarcasm, idioms, or metaphors	TCOV-05-008	Valid partition 1	“Yeah sure, that sounds totally real...”
TCOV-05-009	News containing strong misinformation indicators	TCOV-05-009	Invalid partition 2	False medical or conspiracy claims

Table 2.2.5.1b News Content Equivalence Partitioning Test Conditions & Coverage

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2.2.5.2 Boundary Value Analysis

Boundary Value Analysis is used to test the boundary values of news content length for the AI-Based News Classification feature.

An assumption is made that valid news content must contain a minimum number of characters and must not exceed the maximum allowed length.

Boundary Value Assumptions

- Minimum length of news content = 1 character
- Maximum length of news content = 10,000 characters

Equivalence Partitions

- EP1: $0 \leq \text{News Content} < 1$ character
- EP2: $1 \text{ character} \leq \text{News Content} \leq 10,000$ characters
- EP3: $\text{News Content} > 10,000$ characters

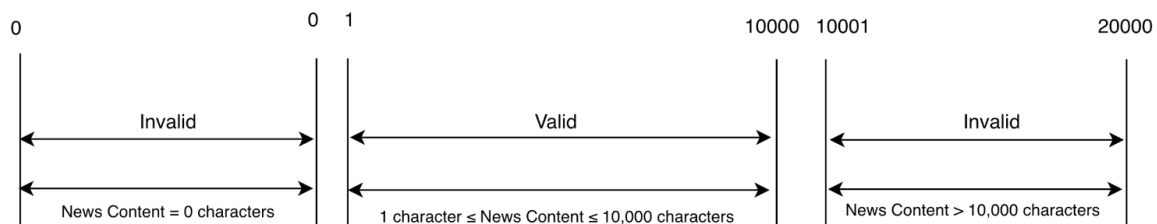


Figure 2.2.5.2a News Content Boundary Value Analysis

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Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-05-010	News Content = 0 character	TCOV-05-010	News Content = 0 character (Invalid)	No input
TCON-05-011	News Content = 1 character	TCOV-05-011	News Content = 1 character (Valid)	“A”
TCON-05-012	News Content = 9,999 characters	TCOV-05-012	News Content = 9,999 characters (Valid)	Long news article text
TCON-05-013	News Content = 10,000 characters	TCOV-05-013	News Content = 10,000 characters (Valid)	Maximum length article
TCON-05-014	News Content = 10,001 characters	TCOV-05-014	News Content = 10,001 characters (Invalid)	Article exceeding limit
TCON-05-015	News Content = 20,000 characters	TCOV-05-015	News Content > maximum length (Invalid)	Extremely long article
TCON-05-016	Valid News Content	TCOV-05-016	Valid News Content (Valid output)	“Health ministry denies the rumour.”
TCON-05-017	Invalid News Content	TCOV-05-017	Invalid News Content (Invalid output)	Empty or unreadable text

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2.2.5.3 Decision Table Testing

Decision table testing is used to test valid combinations of news content characteristics and AI classification outcomes.

Test Condition ID	Test Condition
TCOV-05-002	Valid readable news content
TCOV-05-004	Rephrased content similar to known fake news
TCOV-05-007	Adversarially manipulated news content
TCOV-05-008	News content containing sarcasm or idioms
TCOV-05-009	News containing strong misinformation indicators
TCOV-05-010	No misinformation indicators detected

Condition	TCOV-05-002	TCOV-05-004	TCOV-05-007	TCOV-05-008	TCOV-05-009	TCOV-05-010
Valid readable news content	Y	Y	Y	Y	Y	Y
Rephrased fake news detected	–	Y	–	–	–	N

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Adversarial manipulation detected	–	–	Y	–	–	N
Sarcasm / idioms detected	–	–	–	Y	–	N
Strong misinformation indicators	–	–	–	–	Y	N
Output	True / Unverified	False	Suspicious / Unverified	Misleading	False	True

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Date: 4 January 2026	Test Design Specification ID: SeBenarnya_TDS_1.0.0	

2.2.5.4 State Transition Testing

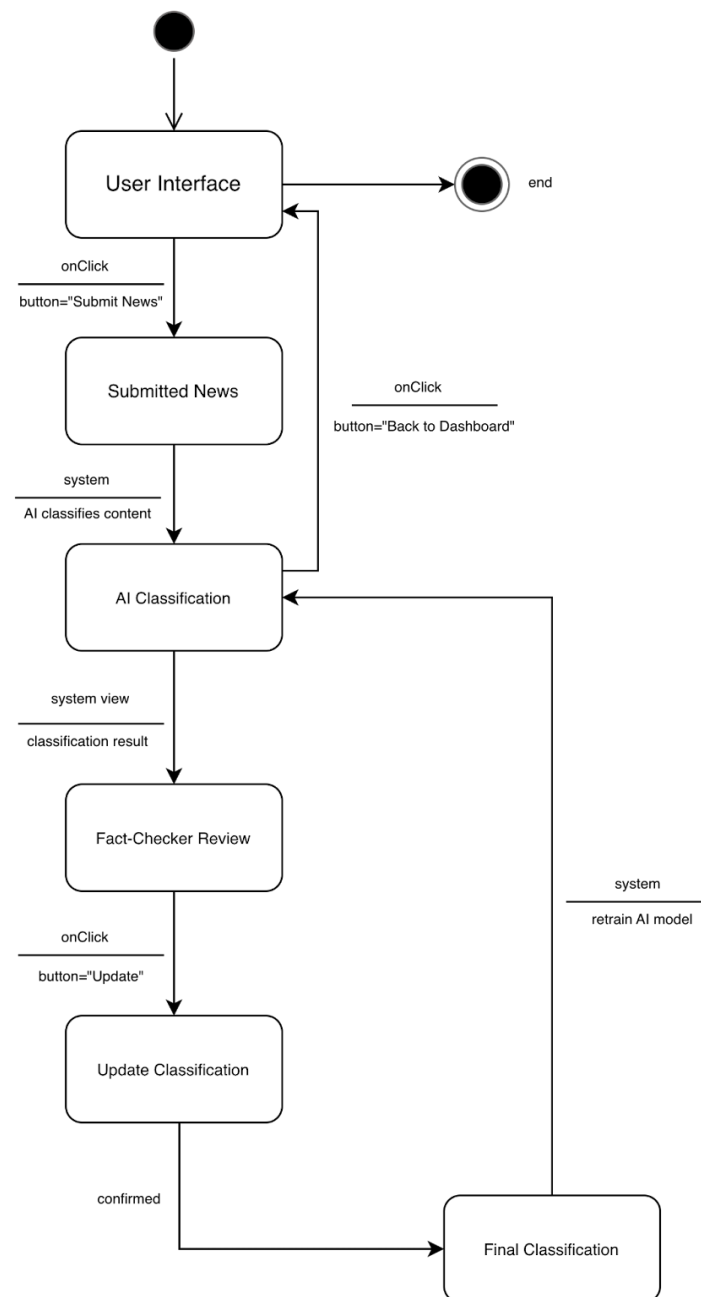


Figure 2.2.5.4a shows the AI-Based News Classification State Diagram

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The test condition for state transition testing is as follows:

- TCON-05-019 AI-Based News Classification State Transition

State / Input	Submit News	AI Classify	Fact-Checker Review	Update Classification	Confirmed
Submitted	S1 / TCOV-05-019	–	–	–	–
Classified	–	S2 / TCOV-05-020	–	–	–
Reviewed	–	–	S3 / TCOV-05-021	–	–
Updated	–	–	–	S4 / TCOV-05-022	–
Classified (Final)	–	–	–	–	S5 / TCOV-05-023

Table 2.2.5.4a AI-Based News Classification State Table

State Definitions

- S1: News content submitted by user
- S2: News content classified by AI
- S3: Classification reviewed by fact-checker
- S4: Classification updated by fact-checker
- S5: Final classification confirmed and stored

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State Transition Coverage

Test Coverage ID	Test Coverage
TCOV-05-019	S1 to S2 with Submit News
TCOV-05-020	S2 to S3 with AI Classification
TCOV-05-021	S3 to S4 with Fact-Checker Review
TCOV-05-022	S4 to S5 with Update Classification
TCOV-05-023	S5 to S2 with Confirmed (AI retraining triggered)

Table 2.2.5.4b AI-Based News Classification State Coverage

2.2.2.5 Use Case Testing

Use case testing is used to test the main flow, alternative flow, and exception flow for AI-Based News Classification.

AI-Based News Classification [REQ-F005-001]

Use Case	UC005	Version	1.0
Feature	F005		
Purpose	To allow the system to classify submitted news content using AI and allow fact-checkers to review and update the classification.		

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Actor	User, Fact-Checker
Trigger	User submits news content for verification.
Precondition	User must submit readable news content. Fact-checker must be logged into the system with appropriate access rights.
Main Flow	8. User submits news content for classification. 9. System validates the submitted news content. 10. System performs AI-based classification of the news content. 11. System displays the classification result (True, False, Misleading, or Unverified). 12. Fact-checker reviews the AI classification result. 13. Fact-checker confirms or updates the classification. 14. System updates the final classification result and stores it. 15. Use case ends.

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Alternative Flow	None.
Exception Flow	E1: Invalid News Content [REQ-F005-002] 1. The system detects that the submitted news content is empty or unreadable. 2. The system displays an error message indicating invalid news content. 3. The use case ends.
Rules	● R1: News content must contain readable and meaningful text. ● R2: The system must flag manipulated or misleading content appropriately. ● R3: Only fact-checkers are allowed to update AI classification results.
Note	None
Author	Nur Liyana Athirah Binti Zailani

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Design Specification ID: SeBenarnya_TDS_1.0.0	

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-05-018	Main Flow	TCOV-05-018	Main flow	Submit valid news content and confirm classification
TCON-05-019	Exception Flow – Invalid News Content	TCOV-05-019	Exception flow – Invalid content	Submit empty news content
TCON-05-020	Alternative Flow – Fact-Checker Updates Classification	TCOV-05-020	Alternative flow – Update classification	Fact-checker updates classification result

Table 2.2.5.5a AI-Based News Classification Use Case Test Conditions & Coverage

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2.2.6 F006 Multi-Language Reporting

Five techniques are identified for designing tests for this feature, including:

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing
5. Use Case Testing

2.2.6.1 Equivalence Partitioning for Report Language Selection

Input of Detected Language

Invalid Partition 1	Valid Partition 1	Valid Partition 2	Valid Partition 3	Invalid Partition 2
No language preference set	Malay language report	English language report	Mandarin language report	Unsupported language request

Table 2.2.6.1a Multi-Language Reporting Equivalence Partitions

Equivalent Partitioning of Report Language = { none, Malay, English, Mandarin, unsupported }

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCOV-06-00 1	User has no language preference	TCOV-06-00 1	Invalid partition 1	preference = null
TCOV-06-00 2	Report generated in Malay	TCOV-06-00 2	Valid partition 1	preference = "ms"

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TCOV-06-003	Report generated in English	TCOV-06-003	Valid partition 2	preference = "en"
TCOV-06-004	Report generated in Mandarin	TCOV-06-004	Valid partition 3	preference = "zh"
TCOV-06-005	User requests unsupported language	TCOV-06-005	Invalid partition 2	preference = "fr"
TCOV-06-006	Report with technical terms translated	TCOV-06-006	Valid partition 1	Classification terms in Malay
TCOV-06-007	Report with statistical data formatted	TCOV-06-007	Valid partition 2	Numbers formatted for English locale

Table 2.2.6.1b Multi-Language Reporting Equivalence Partitioning Test Conditions & Coverage

2.2.6.2 Boundary Value Analysis

Boundary Value Analysis for report content completeness.

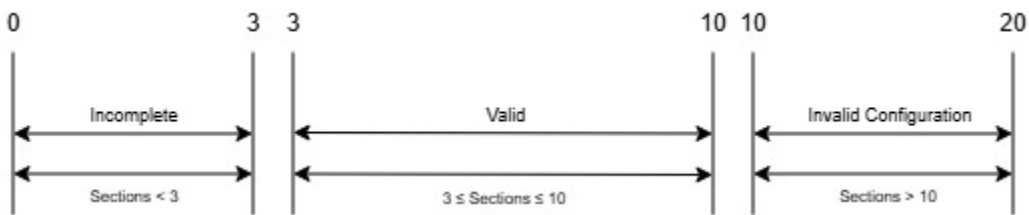
Boundary Value Assumptions

- Minimum report sections = 3 (Header, Classification Result, Footer)
- Maximum report sections = 10 (all available sections)

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Date: 4 January 2026	Test Design Specification ID: SeBenarnya_TDS_1.0.0	

Equivalence Partitions

- EP1: Sections < 3 (incomplete)
- EP2: $3 \leq \text{Sections} \leq 10$ (valid)
- EP3: Sections > 10 (invalid configuration)



Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-06-008	Report with 0 sections	TCOV-06-008	Incomplete (Invalid)	Empty report
TCON-06-009	Report with 2 sections	TCOV-06-009	Below minimum (Invalid)	Missing required sections
TCON-06-010	Report with 3 sections	TCOV-06-010	Minimum valid (Valid)	Basic complete report
TCON-06-011	Report with 6 sections	TCOV-06-011	Standard report (Valid)	Full detailed report
TCON-06-012	Report with 10 sections	TCOV-06-012	Maximum sections (Valid)	Comprehensive report

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2.2.6.3 Decision Table Testing

Condition	TCOV-06-002	TCOV-06-003	TCOV-06-004	TCOV-06-007	TCOV-06-001
Language preference valid	Y	Y	Y	Y	N
Classification data available	Y	Y	Y	Y	Y
Report template exists	Y	Y	Y	-	Y
User authenticated	Y	Y	Y	Y	Y
Output	Generate Malay Report	Generate English Report	Generate Mandarin Report	Error: Unsupported Language	Generate Default Language Report

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2.2.6.4 State Transition Testing

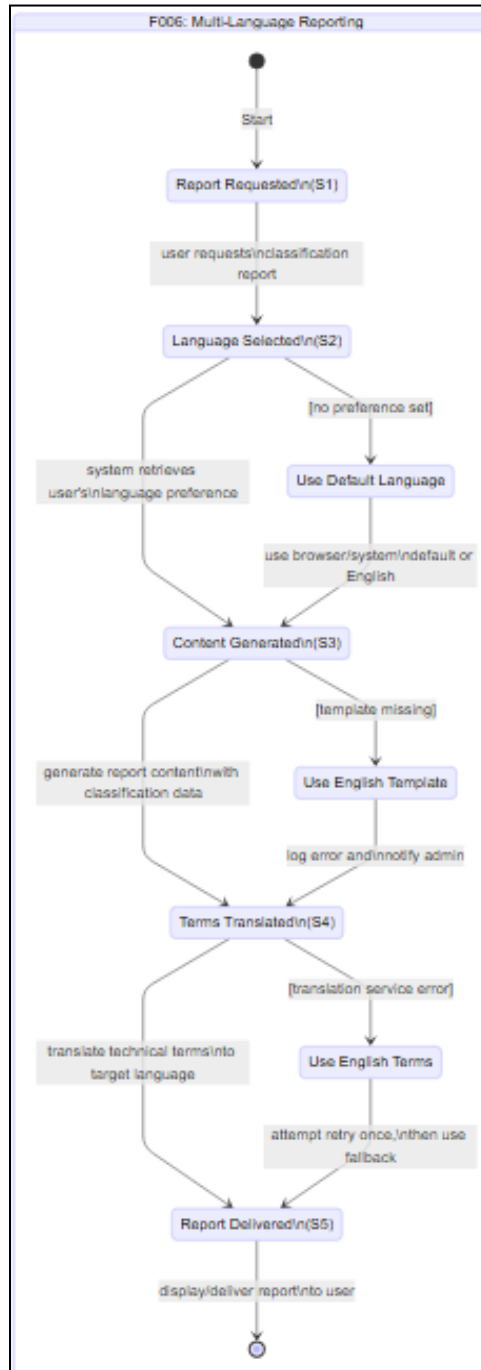


Figure 2.2.6.4a State Diagram

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State Table

State/Input	Request Report	Select Language	Generate Content	Translate Terms	Deliver Report
Report Requested	S1/TCOV-06-013	-	-	-	
Language Selected	-	S2/TCOV-06-014	-	-	
Content Generated	-	-	S3/TCOV-06-015	-	
Terms Translated	-	-	-	S4/TCOV-06-016	
Report Delivered	-	-	-	-	S5/TCOV-06-017

State Definitions:

- S1: User requests classification report
- S2: System identifies user's language preference
- S3: System generates report content with classification data
- S4: System translates technical terms and labels to target language
- S5: Report delivered to user in selected language

State Transition Coverage:

Test Coverage ID	Test Coverage
TCOV-06-013	S1 to S2 with Request Report
TCOV-06-014	S2 to S3 with Select Language
TCOV-06-015	S3 to S4 with Generate Content

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TCOV-06-016	S4 to S5 with Translate Terms
TCOV-06-017	S5 completion with Deliver Report

2.2.6.5 Use Case Testing

Multi-Language Reporting [REQ-02-005]

Use Case	UC006	Version	1.0
Feature	F006		
Purpose	To generate classification reports in the user's preferred language with accurate terminology		
Actor	User, System		
Trigger	User requests classification report or system automatically generates report after classification		
Precondition	Classification result must be available; User must have language preference set or system default language available		
Main Flow	1. User requests classification report for a news submission 2. System retrieves user's language preference from profile 3. System validates language preference is supported (Malay/English/Mandarin) 4. System retrieves classification data and metadata		

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	5. System selects appropriate language-specific report template 6. System translates technical terms and labels to target language 7. System formats dates, numbers, and statistics according to locale 8. System generates complete report in selected language 9. System displays or delivers report to user 10. Use case ends
Alternative Flow	A1: User Changes Language Preference 1. At step 2, user selects different language before report generation 2. System updates language preference 3. Resume at step 3 with new language preference
Exception Flow	E1: No Language Preference Set [REQ-02-005] 1. At step 2, system finds no language preference 2. System uses browser/system default language 3. If default not supported, system uses English as fallback 4. Resume at step 4 E2: Report Template Missing [REQ-02-005] 1. At step 5, system cannot find language-specific template 2. System logs error and notifies administrator 3. System uses English template as fallback

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	4. System displays warning to user about language limitation 5. Resume at step 6 E3: Translation Service Error [REQ-02-005] 1. At step 6, translation service fails 2. System attempts retry once 3. If retry fails, system uses English terms with language note 4. System logs error and notifies administrator 5. Resume at step 8
Rules	● R1: All supported languages (Malay, English, Mandarin) must have complete report templates ● R2: Technical terminology must maintain accuracy across all languages ● R3: Report must include: classification result, confidence score (if authorized), submission details, and timestamp ● R4: Date and number formatting must follow target language locale conventions ● R5: System must log language preference for future use
Note	Reports must maintain consistent formatting and information completeness across all languages.
Author	Nur Izzah Dania binti Hipni Shahlizal

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Use Case Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-06-018	Main Flow	TCOV-06-018	Main flow	Generate report in Malay for verified news
TCON-06-019	Alternative Flow - Change Language	TCOV-06-019	Alternative flow	User switches from English to Mandarin
TCON-06-020	Exception - No Preference	TCOV-06-020	Exception flow	New user with no language preference
TCON-06-021	Exception - Template Missing	TCOV-06-021	Exception flow	Request report when Mandarin template unavailable
TCON-06-022	Exception - Translation Error	TCOV-06-022	Exception flow	Generate report during translation service outage

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2.2.7 F007 User Submission

Five techniques are identified for designing test for this feature, including:

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Use Case Testing

2.2.7.1 Equivalence Partitioning for User Submission

Equivalence partitioning is used to split user submission input into possible partitions for testing.

Equivalent Partitions

User Submission = { empty submission, valid readable submission, invalid / unreadable submission }

Table 2.2.7.1a User Submission Equivalence Partitions

Invalid Partition 1	Valid Partition 1	Invalid Partition 2
Empty submission	Readable and meaningful submission	Unreadable or meaningless submission

User Submission Equivalence Partitioning Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-07-001	Submission = empty	TCOV-07-001	Invalid partition 1	Empty input
TCON-07-002	Valid readable submission	TCOV-07-002	Valid partition 1	“Health ministry denies the rumour.”

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TCON-07-003	Unreadable or random input	TCOV-07-003	Invalid partition 2	“@@@####\$”
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2.2.7.2 Boundary Value Analysis for User Submission

Boundary Value Analysis is used to test the boundary values of user submission length.

Boundary Value Assumptions

- Minimum submission length = 1 character
- Maximum submission length = 10,000 characters

Equivalence Partitions

- EP1: $0 \leq \text{Submission} < 1$ character
- EP2: $1 \leq \text{Submission} \leq 10,000$ characters
- EP3: $\text{Submission} > 10,000$ characters

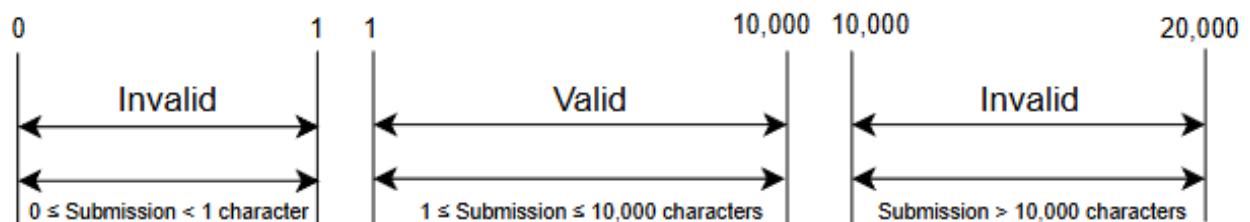


Table 2.2.7.2a User Submission Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-07-004	Submission = 0 character	TCOV-07-004	Invalid	No input
TCON-07-005	Submission = 1 character	TCOV-07-005	Valid	“A”

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TCOV-07-006	Submission = 10,000 characters	TCOV-07-006	Valid	Maximum length text
TCOV-07-007	Submission > 10,000 characters	TCOV-07-007	Invalid	Over-limit text

2.2.7.3 Decision Table Testing

Decision table testing is used to test valid combinations of submission validity and system response.

Test Conditions

- C1: Submission readable
- C2: Submission unreadable

Condition	TCOV-07-002	TCOV-07-003
Submission readable	Y	N
Submission unreadable	N	Y
Output	Submission stored	Submission rejected

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2.2.7.4 State Transition Testing

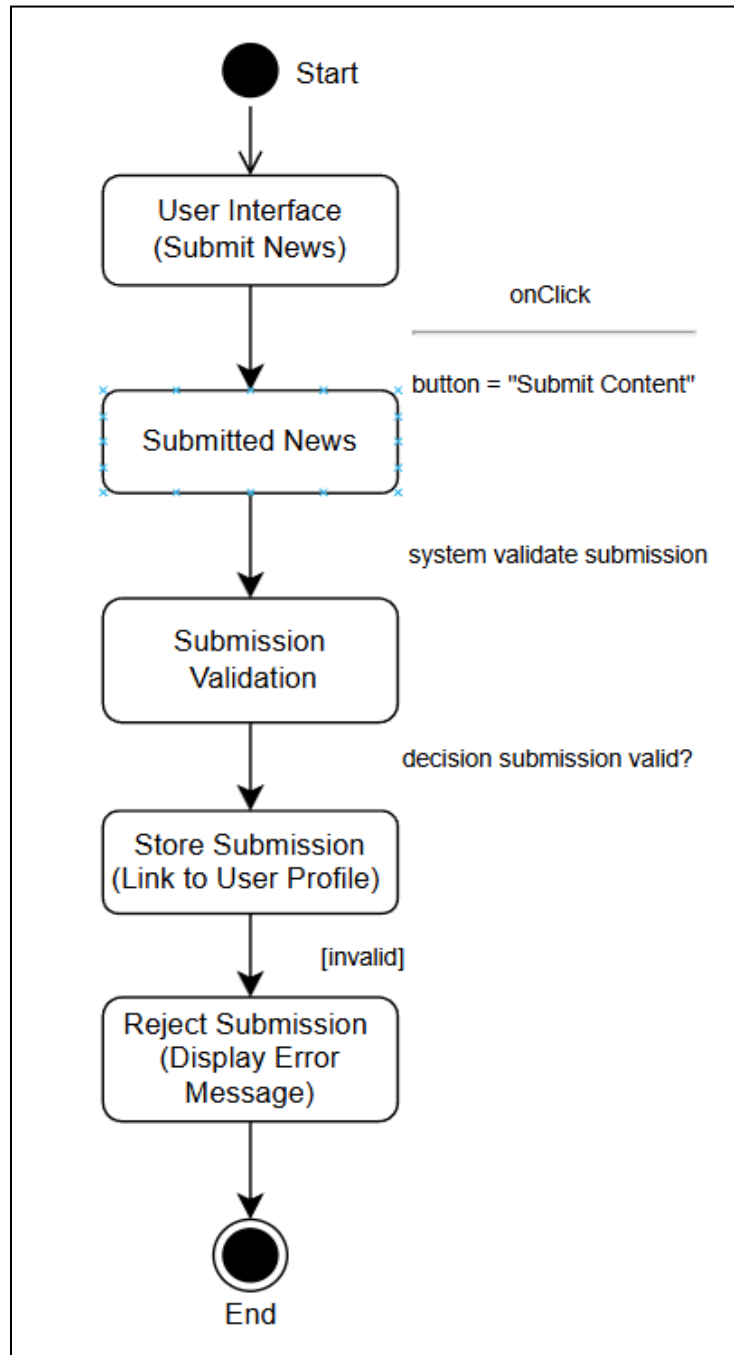


Figure 2.2.7.4a shows the User Submission State Diagram.

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State transition testing is used to verify that user submission flows correctly from submission to validation and finally to either stored or rejected state.

State Definitions

- S1: User prepares submission
- S2: Submission submitted
- S3: Submission validated
- S4: Submission stored
- S5: Submission rejected

Table 2.2.7.4a User Submission State Table

State / Input	Submit News	Validate Submission	Store / Reject
Prepared	S1 / TCOV-07-009	–	–
Submitted	–	S2 / TCOV-07-010	–
Validated	–	–	S3 / TCOV-07-011
Rejected	–	–	S4 / TCOV-07-012

State Transition Coverage

Test Coverage ID	Test Coverage
TCOV-07-009	S1 to S2 with Submit News
TCOV-07-010	S2 to S3 with Validation
TCOV-07-011	S3 to S4 with Valid Submission
TCOV-07-012	S3 to S5 with Invalid Submission

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2.2.7.5 Use Case Testing

Use case testing is used to test the main flow and exception flow for User Submission.

User Submission [REQ-03-001]

Use Case	UC007	Version	1.0
Feature	F007		
Purpose	To allow users to submit news content into the system for verification and further processing.		
Actor	User		
Trigger	User submits news content through the submission interface.		
Precondition	User must be logged into the system.		
Main Flow	1. User enters news content into the submission form. 2. User submits the news content to the system. 3. The system validates the submitted news content. 4. The system stores the submission and links it to the user's account profile. 5. Use case ends.		
Alternative Flow	None.		

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Exception Flow	E1: Invalid News Content [REQ-03-001] 1. The system detects that the submitted news content is empty or unreadable. 2. The system displays an error message indicating invalid news content. 3. The use case ends.
Rules	• R1: User submission must contain readable and meaningful text. • R2: Empty or unreadable submissions shall not be stored in the system.
Note	None.
Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCOV-07-018	Main Flow	TCOV-07-018	Main flow	Submit valid and readable news content
TCOV-07-019	Exception Flow – Invalid News Content	TCOV-07-019	Exception flow – Invalid content	Submit empty or unreadable news content

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TCOV-07-020	Alternative Flow – Submission Rejected	TCOV-07-020	Alternative flow – Reject submission	Submit news content exceeding length limit
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Table 2.2.7.5a User Submission Use Case Test Conditions & Coverage

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2.2.8 F008 Fact-Checker Verification Workflow

Five techniques are identified for designing test for this feature, including:

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Use Case Testing

2.2.8.1 Equivalence Partitioning for Fact-Checker Verification

Equivalence partitioning is used to split fact-checker verification outcomes into possible partitions for testing.

Equivalent Partitioning of Verification Outcome

= { verified true, verified false, not verified }

Verification Outcome

Valid Partition 1	Valid Partition 2	Invalid Partition
Verified as True	Verified as False	Not verified / unauthorized

Table 2.2.8.1a Fact-Checker Verification Equivalence Partitioning Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-08-001	Submission verified as True	TCOV-08-001	Valid partition 1	Fact-checker confirms true news
TCON-08-002	Submission verified as False	TCOV-08-002	Valid partition 2	Fact-checker confirms false news

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TCOV-08-003	Submission not verified	TCOV-08-003	Invalid partition	Verification skipped or unauthorized
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2.2.8.2 Boundary Value Analysis

Boundary Value Analysis is used to test system behaviour when a user approaches or exceeds violation thresholds.

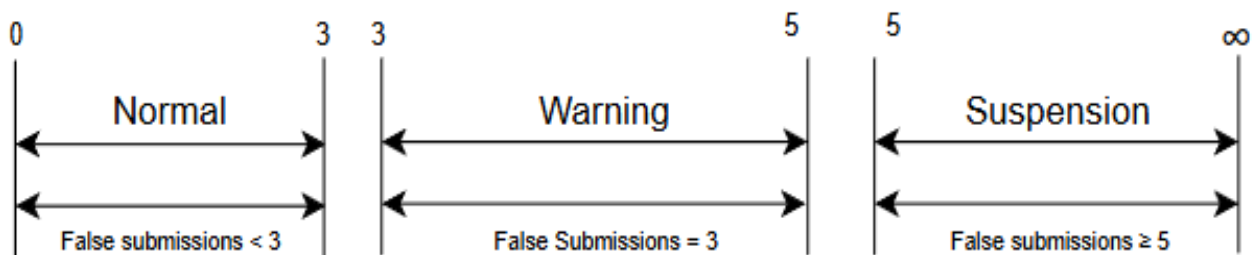
An assumption is made that enforcement actions are triggered when a certain number of false submissions is reached.

Boundary Value Assumptions

- Warning threshold = 3 false submissions
- Suspension threshold = 5 false submissions

Equivalence Partitions

- EP1: False submissions < 3
- EP2: False submissions $= 3$
- EP3: False submissions ≥ 5



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Table 2.2.8.2a False Submission Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-08-004	False count = 2	TCOV-08-004	Below threshold	User false count = 2
TCON-08-005	False count = 3	TCOV-08-005	Warning threshold	User false count = 3
TCON-08-006	False count = 5	TCOV-08-006	Suspension threshold	User false count = 5

2.2.8.3 Decision Table Testing

Decision table testing is used to test valid combinations of verification results and system enforcement actions.

Conditions

- C1: Submission verified as False
- C2: Warning threshold reached
- C3: Suspension threshold reached

Table 2.2.8.3a Enforcement Decision Table

Condition / Rule	R1	R2	R3
Verified as False	Y	Y	Y
Warning threshold reached	N	Y	Y
Suspension threshold reached	N	N	Y
Action	Update reputation	Issue warning	Suspend user

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2.2.8.4 State Transition Testing

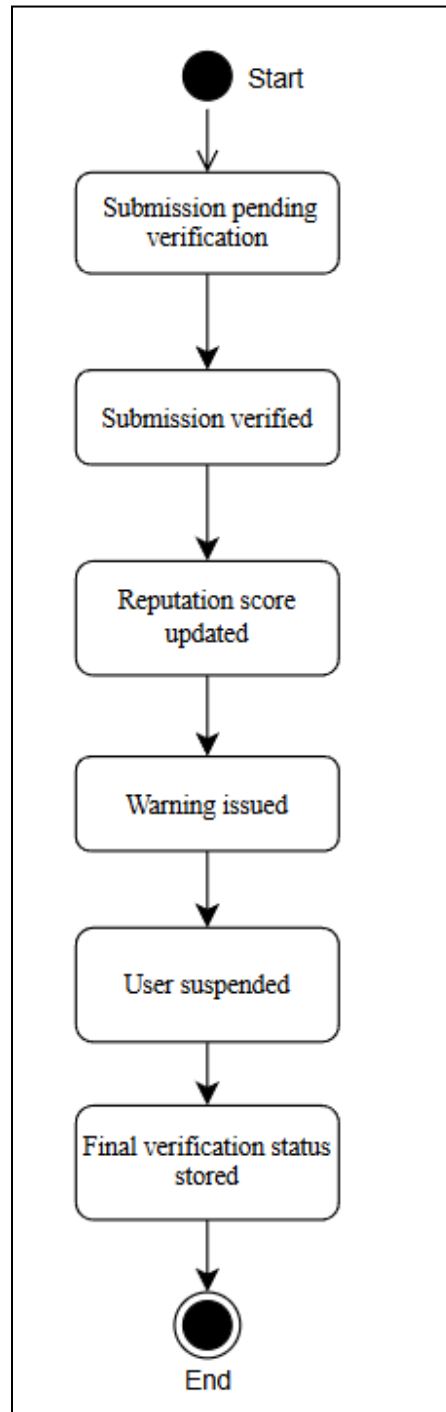


Figure 2.2.8.4a shows the Fact-Checker Verification Workflow State Diagram.

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State transition testing is used to verify that verified submissions correctly transition through reputation update and enforcement states.

State Definitions

- S1: Submission pending verification
- S2: Submission verified
- S3: Reputation score updated
- S4: Warning issued
- S5: User suspended
- S6: Final verification status stored

Table 2.2.8.4a Fact-Checker Verification State Table

State / Input	Verify Submission	Update Reputation	Enforce Action
Pending	S1 / TCOV-08-007	–	–
Verified	–	S2 / TCOV-08-008	–
Reputation Updated	–	–	S3 / TCOV-08-009
Warning Issued	–	–	S4 / TCOV-08-010
Suspended	–	–	S5 / TCOV-08-011

State Transition Coverage

Test Coverage ID	Test Coverage
TCOV-08-007	S1 to S2 with Verification
TCOV-08-008	S2 to S3 with Reputation Update
TCOV-08-009	S3 to S4 when Warning Threshold Reached
TCOV-08-010	S4 to S5 when Suspension Threshold Reached
TCOV-08-011	S3 to S6 when No Enforcement Needed

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2.2.8.5 Use Case Testing

Use case testing is used to test the main flow, alternative flow, and exception flow for the fact-checker verification workflow.

Fact-Checker Verification Workflow [REQ-03-002 – REQ-03-005]

Use Case	UC008	Version	1.0
Feature	F008		
Purpose	To allow fact-checkers to verify user submissions and trigger reputation and enforcement actions.		
Actor	Fact-Checker		
Trigger	Fact-checker reviews a submitted news item.		
Precondition	Submission exists and fact-checker is logged in with proper access rights.		
Main Flow	1. Fact-checker reviews submitted news content. 2. Fact-checker verifies the submission as True or False. 3. The system updates the false submission count if applicable. 4. The system updates the user's reputation score. 5. The system issues warning or suspension if threshold is reached. 6. The system sends an automated alert.		

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	7. Use case ends.
Alternative Flow	A1: Fact-Checker Updates Classification - Fact-checker updates the verification result. - The system recalculates reputation and enforcement actions.
Exception Flow	E1: Unauthorized Access - System denies verification access. - Use case ends.
Rules	R1: Only fact-checkers are allowed to verify submissions. R2: Only verified false submissions affect reputation and enforcement. R3: Alerts must be sent before suspension.
Note	None.
Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh

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Table 2.2.8.5a Fact-Checker Verification Workflow Use Case Test Conditions & Coverage

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-08-018	Main Flow	TCOV-08-018	Main flow	Fact-checker reviews submission and confirms verification result
TCON-08-019	Exception Flow – Unauthorized Access	TCOV-08-019	Exception flow – Access denied	Unauthorized user attempts to verify submission
TCON-08-020	Alternative Flow – Update Classification	TCOV-08-020	Alternative flow – Update verification result	Fact-checker updates the classification from AI result

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2.3 Test Identification

2.3.1 Multi-Language Detection

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-01-001	Verify detection of pure Malay content	TCOV-01-002	TCON-01-002
TC-01-002	Verify detection of pure English content	TCOV-01-003	TCON-01-003
TC-01-003	Verify detection of pure Mandarin content	TCOV-01-004	TCON-01-004
TC-01-004	Verify rejection of empty content	TCOV-01-001	TCON-01-001
TC-01-005	Verify handling of mixed language content	TCOV-01-005	TCON-01-005
TC-01-006	Verify minimum character threshold	TCOV-01-011	TCON-01-011
TC-01-007	Verify maximum character limit	TCOV-01-013	TCON-01-013
TC-01-008	Verify detection with proper nouns in text	TCOV-01-008	TCON-01-008

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2.3.2 Language Routing to Appropriate Classifier

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-02-001	Verify routing of Malay content to Malay classifier	TCOV-02-002	TCON-02-002
TC-02-002	Verify routing of English content to English classifier	TCOV-02-003	TCON-02-003
TC-02-003	Verify routing of Mandarin content to Mandarin classifier	TCOV-02-004	TCON-02-004
TC-02-004	Verify handling of unsupported language	TCOV-02-005	TCON-02-005
TC-02-005	Verify routing with minimum confidence threshold	TCOV-02-010	TCON-02-010
TC-02-006	Verify routing rejection below confidence threshold	TCOV-02-009	TCON-02-009
TC-02-007	Verify handling of classifier unavailability	TCOV-02-007	TCON-02-007

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2.3.3 Cross-Language Duplicate Detection

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-03-001	Verify detection of exact duplicate in same language	TCOV-03-003	TCON-03-003
TC-03-002	Verify detection of semantic duplicate across languages	TCOV-03-004	TCON-03-004
TC-03-003	Verify unique content proceeds to classification	TCOV-03-002	TCON-03-002
TC-03-004	Verify paraphrased content detection	TCOV-03-006	TCON-03-006
TC-03-005	Verify similarity threshold at 80%	TCOV-03-010	TCON-03-010
TC-03-006	Verify content below similarity threshold proceeds	TCOV-03-009	TCON-03-009
TC-03-007	Verify handling of database search failure	TCOV-03-020	TCON-03-020

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2.3.4 Translation Validation

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-04-001	Verify approval of high-quality translation	TCOV-04-002	TCON-04-002
TC-04-002	Verify approval of acceptable quality translation	TCOV-04-003	TCON-04-003
TC-04-003	Verify rejection of poor quality translation	TCOV-04-004	TCON-04-004
TC-04-004	Verify rejection of translation with factual errors	TCOV-04-007	TCON-04-007
TC-04-005	Verify translation quality threshold at 75%	TCOV-04-010	TCON-04-010
TC-04-006	Verify rejection below quality threshold	TCOV-04-009	TCON-04-009
TC-04-007	Verify preservation of technical terms	TCOV-04-005	TCON-04-005
TC-04-008	Verify handling of idioms and phrases	TCOV-04-006	TCON-04-006

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2.3.5 AI-Based News Classification

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-01-001	Verify classification of valid news content	TCOV-05-002	TCON-05-002
		TCOV-05-017	TCON-05-017
TC-01-002	Verify AI model retraining after fact-checker verification	TCOV-05-003	TCON-05-003
TC-01-003	Verify detection of rephrased fake news patterns	TCOV-05-004	TCON-05-004
TC-01-004	Verify generation of explainable AI results	TCOV-05-005	TCON-05-005
TC-01-005	Verify role-based visibility of confidence score	TCOV-05-006	TCON-05-006
TC-01-006	Verify classification of adversarially manipulated news content	TCOV-05-007	TCON-05-007
TC-01-007	Verify classification of news containing sarcasm or idioms	TCOV-05-008	TCON-05-008
TC-01-008	Verify XAI explanation accuracy for misinformation indicators	TCOV-05-009	TCON-05-009
TC-01-009	Verify system response for empty news content	TCOV-05-001	TCON-05-001

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2.3.6 Multi-Language Reporting

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-06-001	Verify report generation in Malay	TCOV-06-002	TCON-06-002
TC-06-002	Verify report generation in English	TCOV-06-003	TCON-06-003
TC-06-003	Verify report generation in Mandarin	TCOV-06-004	TCON-06-004
TC-06-004	Verify handling of unsupported language request	TCOV-06-005	TCON-06-005
TC-06-005	Verify default language when no preference set	TCOV-06-001	TCON-06-001
TC-06-006	Verify accuracy of translated technical terms	TCOV-06-006	TCON-06-006
TC-06-007	Verify proper locale formatting in reports	TCOV-06-007	TCON-06-007
TC-06-008	Verify report completeness with minimum sections	TCOV-06-010	TCON-06-010

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2.3.7 User Submission

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-07-001	Verify that valid user submissions are successfully accepted and stored	TCOV-07-002	TCON-07-002
TC-07-002	Verify system response when user submits empty news content	TCOV-07-001	TCON-07-001
TC-07-003	Verify system response when user submits unreadable or meaningless content	TCOV-07-003	TCON-07-003
TC-07-004	Verify system behavior when submission length is below minimum limit	TCOV-07-004	TCON-07-004
TC-07-005	Verify system behavior when submission length exceeds maximum limit	TCOV-07-008	TCON-07-008

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2.3.8 Fact-Checker Verification Workflow

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-08-001	Verify fact-checker can verify a submission as true	TCOV-08-001	TCON-08-001
TC-08-002	Verify fact-checker can verify a submission as false	TCOV-08-002	TCON-08-002
TC-08-003	Verify system behaviour when submission is not verified	TCOV-08-003	TCON-08-003
TC-08-004	Verify warning is issued when false submission threshold is reached	TCOV-08-005	TCON-08-005
TC-08-005	Verify user account is suspended after exceeding violation threshold	TCOV-08-006	TCON-08-006
TC-08-006	Verify automated alert is sent when user approaches violation threshold	TCOV-08-005	TCON-08-005
TC-08-007	Verify system denies verification access to unauthorized users	TCOV-08-003	TCON-08-003

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2.4 Feature Pass/Fail Criteria

Each feature must satisfy the following criteria in order to pass:

- I. Pass all its test cases
- II. 0% calculation error if the function involved monetary transaction
- III. No deadlock
- IV. Fulfil its requirement

2.5 Test Deliverables

The following documents will be generated by the system test group and will be delivered during the system testing.

Test documentation:

- i. Test Plan
- ii. Test Design Specifications
- iii. Test Case Specifications
- iv. Test Logs
- v. Test Incident Reports
- vi. Test Summary Report